Locally-Adaptive Spatial Smoothing with Shrinkage-Prior Markov Random Fields

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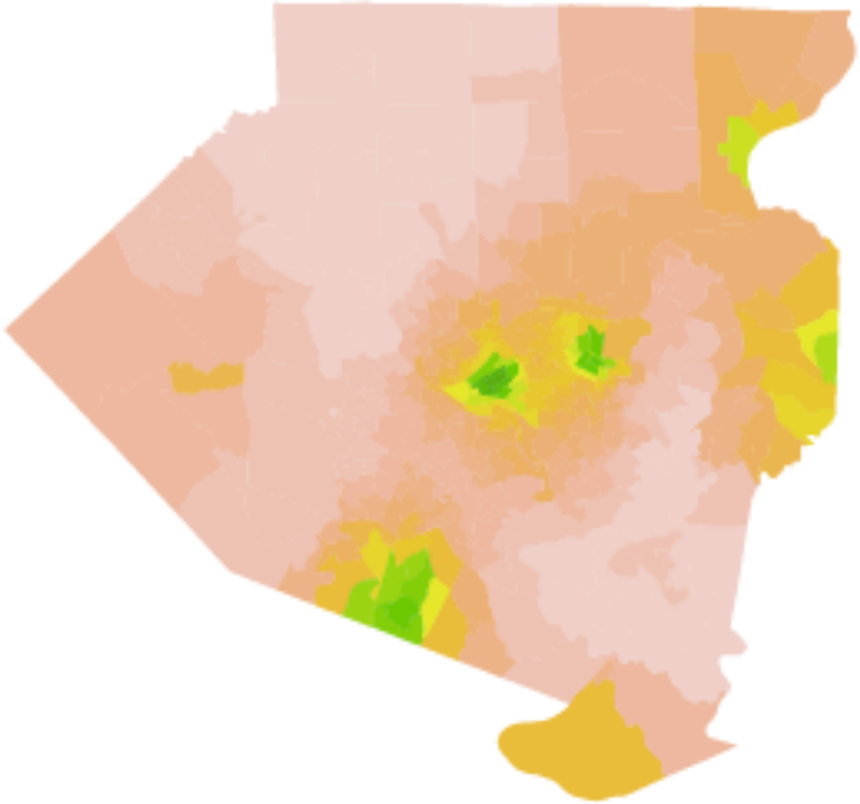
Motivation

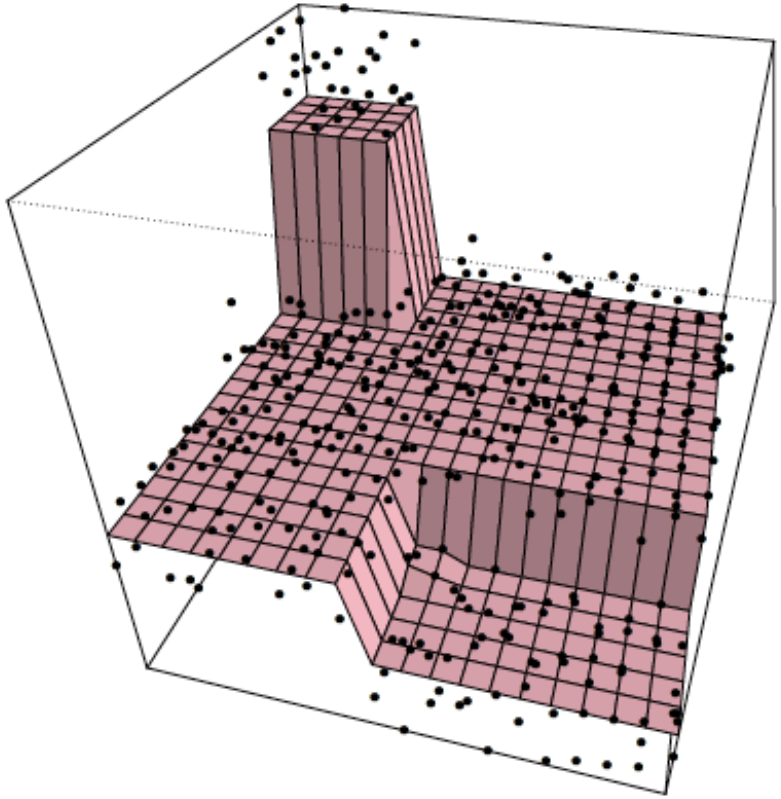
Suppose the following:

- We have spatially referenced data and want to make inference about the underlying spatial process
- The spatial process has breakpoints, sharp local features, or varying smoothness
- We want a fully Bayesian method to estimate the spatial surface that can adapt to local sharp features and still smooth over noisy regions
- We prefer the method be easy to understand and straight forward to implement

Example spatial processes

From Wang et al. 2016





Current Bayesian methods

- Non-stationary Gaussian processes (*e.g.*, Paciorek and Schervish 2006)
- Non-Gaussian processes (*e.g.*, Bolin 2014)
- Adaptive Gaussian Markov random fields (*e.g.* Yue et al. 2010, 2014)

GMRF smoothing prior

Assume there is a process in continuous space that follows an unknown function $f(\boldsymbol{s})$, where $\boldsymbol{s} \in \mathbb{R}^2$.

Let $\theta_i = \tilde{f}(\boldsymbol{a}_i)$ be the expected value of the surface over some discrete areal unit $\boldsymbol{a}_i$

GMRF smoothing prior

Assume there is a process in continuous space that follows an unknown function $f(\boldsymbol{s})$, where $\boldsymbol{s} \in \mathbb{R}^2$.

Let $\theta_i = \tilde{f}(\boldsymbol{a}_i)$ be the expected value of the surface over some discrete areal unit $\boldsymbol{a}_i$

Then a simple kth-order GMRF prior for $\boldsymbol{\theta}$ is induced by letting:

$$\Delta^k \theta_j \sim \mathrm{N}(0, \gamma^2)$$

where $\Delta^k \theta_j$ is a kth-order spatial difference operator defined over the set of J unique pairs of spatial units.

GMRF smoothing prior

Specifically,

$$\boldsymbol{\theta} \mid \mu, \mathbf{Q} \sim \mathrm{N}(\boldsymbol{\mu}, \mathbf{Q}^{-1})$$

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The precision matrix $\boldsymbol{Q}$ or the first-order model is given by,

$$Q_{ij} = \frac{1}{\gamma^2} \begin{cases} w_{i+} & j = i \\ -w_{ij} & j \neq i \\ 0 & \text{otherwise,} \end{cases}$$

where

$$w_{ij} = \frac{1}{d_{ij}}$$

and

$$w_{i+} = \sum_{j \sim i} \frac{1}{d_{ij}}$$

and

$$\gamma \sim \mathrm{C}^{+}(0, \zeta)$$

Adaptive smoothing prior

We can allow locally-adaptive behavior and increase smoothing properties by putting a **shrinkage prior** on $\Delta^k \theta_j$:

$$\Delta^k \theta_j \sim \text{Horseshoe}(0, \gamma)$$

$$\gamma \sim \text{C}^+(0, \zeta)$$

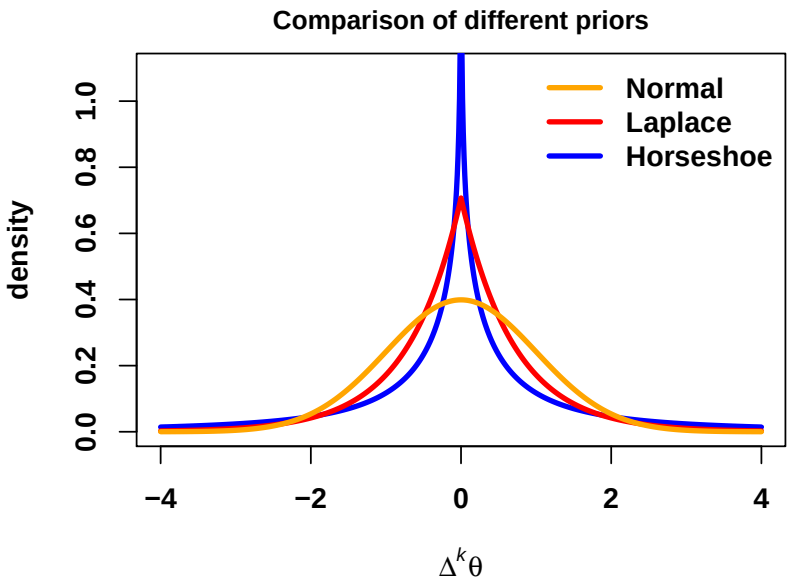
where γ is the global smoothing parameter.

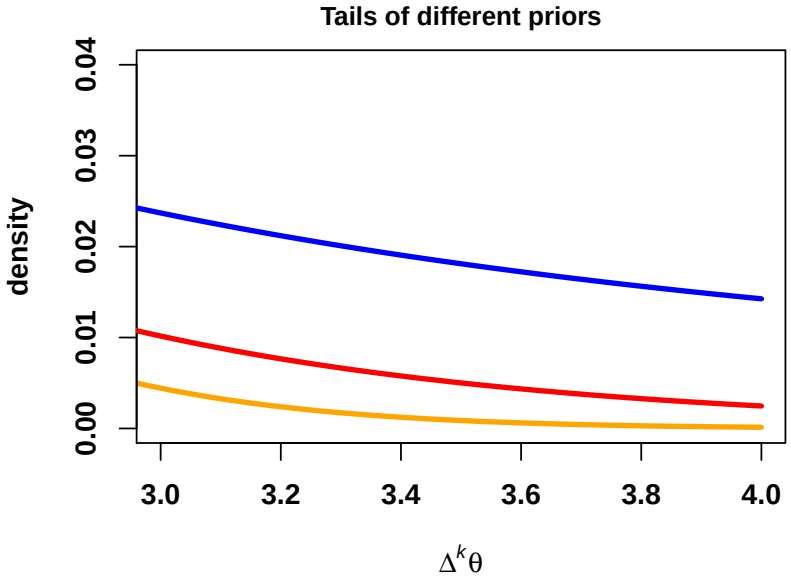
The result is non-Gaussian (horseshoe) Markov random field prior for $\boldsymbol{\theta}$.

Shrinkage priors

A good shrinkage prior will:

- Shrink weak signals: *high mass near zero*
- Let strong signals through: *long tails*





Adaptive smoothing prior

We can represent this horsehoe formulation hierarchically by

$$\boldsymbol{\theta} \mid \mathbf{Q}(\boldsymbol{\tau}) \sim \mathrm{N}(\boldsymbol{\mu}, \mathbf{Q}(\boldsymbol{\tau})^{-1})$$
$$\tau_j \mid \gamma \sim \mathrm{C}^+(0, \gamma)$$
$$\gamma \sim \mathrm{C}^+(0, \zeta)$$

The precision matrix is represented as previously by not including γ and here

$$w_{ij} = \frac{1}{d_{ij}\tau_{ij}^2}$$

Posterior inference

- Hamiltonian Monte Carlo (HMC)
- Uses Hamiltonian dynamics to generate joint parameter proposals

- Pro: results in high acceptance rates
- Con: can be slow due to cost of gradient calculations

- Sparse matrix operations can reduce computation costs

Sparse computations

Let the $(J+1) \times N$ matrix $\mathbf{B}$ consist of differencing operators for the set of J unique pairwise differences between adjacent map units, such that

$$\mathbf{B}\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \Delta\theta_1 \\ \vdots \\ \Delta\theta_J \end{bmatrix} = \mathbf{z}$$

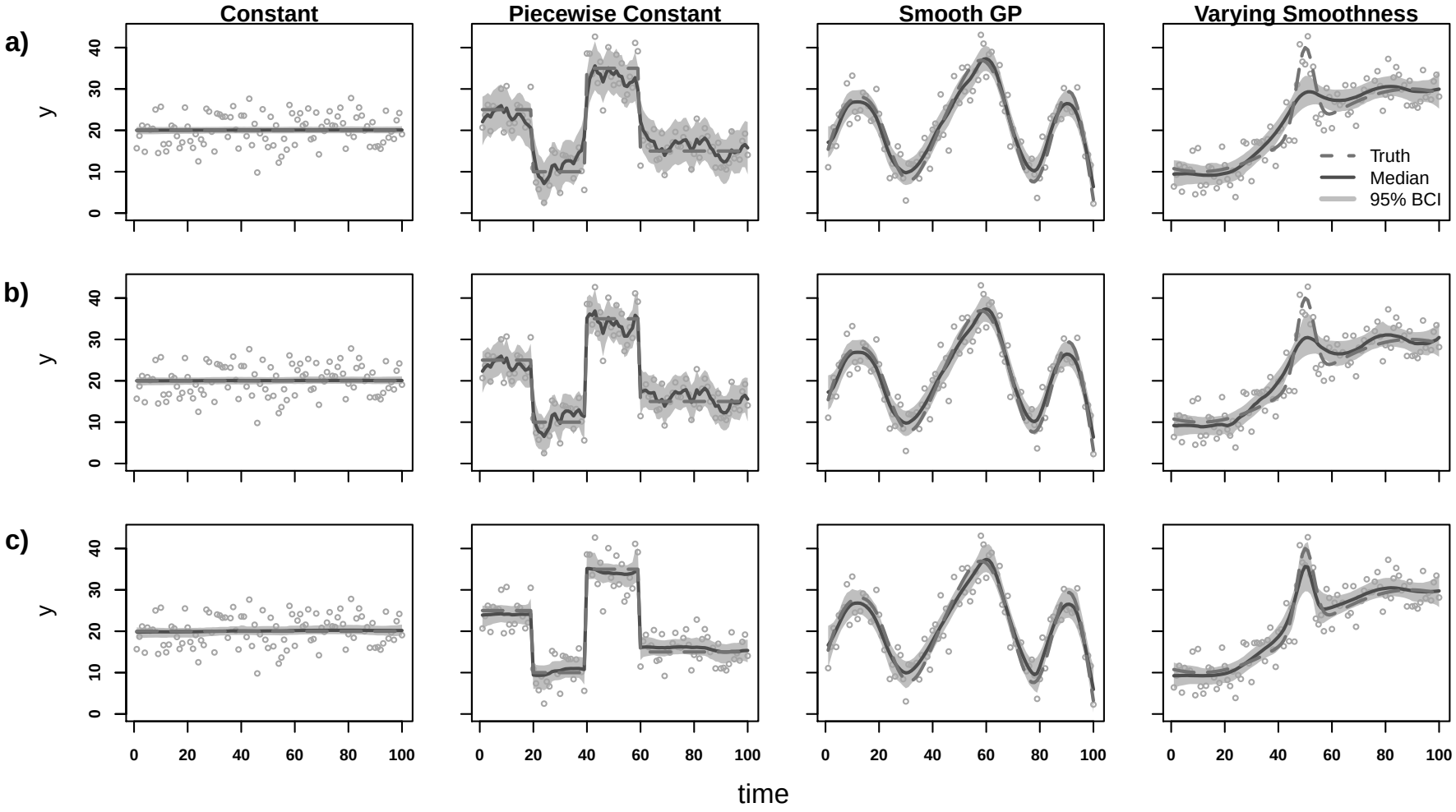
Note that the first row of $\mathbf{B}$ has a 1 in the first element and 0's in the remaining elements. The remaining rows have a single 1 and -1 and the rest 0's.

Sparse computations

- For computational efficiency, want to sample $\mathbf{z}$ instead of $\boldsymbol{\theta}$.
- Need $\boldsymbol{\theta}$ for likelihood computations.
- Need to solve for $\boldsymbol{\theta}$ using $\mathbf{B}$ and $\mathbf{z}$.
- $\mathbf{B}$ is rectangular but can be decomposed with a QR decomposition.
- Let $\mathbf{Q}_B\mathbf{R}_B = \mathbf{B}$,
- then $\boldsymbol{\theta} = \mathbf{R}_B^{-1}\mathbf{Q}_B^T\mathbf{z}$
- Note $\mathbf{R}_B^{-1}$ and $\mathbf{Q}_B^T$ are partially sparse and only need to be computed once and saved.

Results from 1-D

a) GMRF ; b) Laplace RF ; c) Horseshoe RF (Faulkner and Minin 2017)



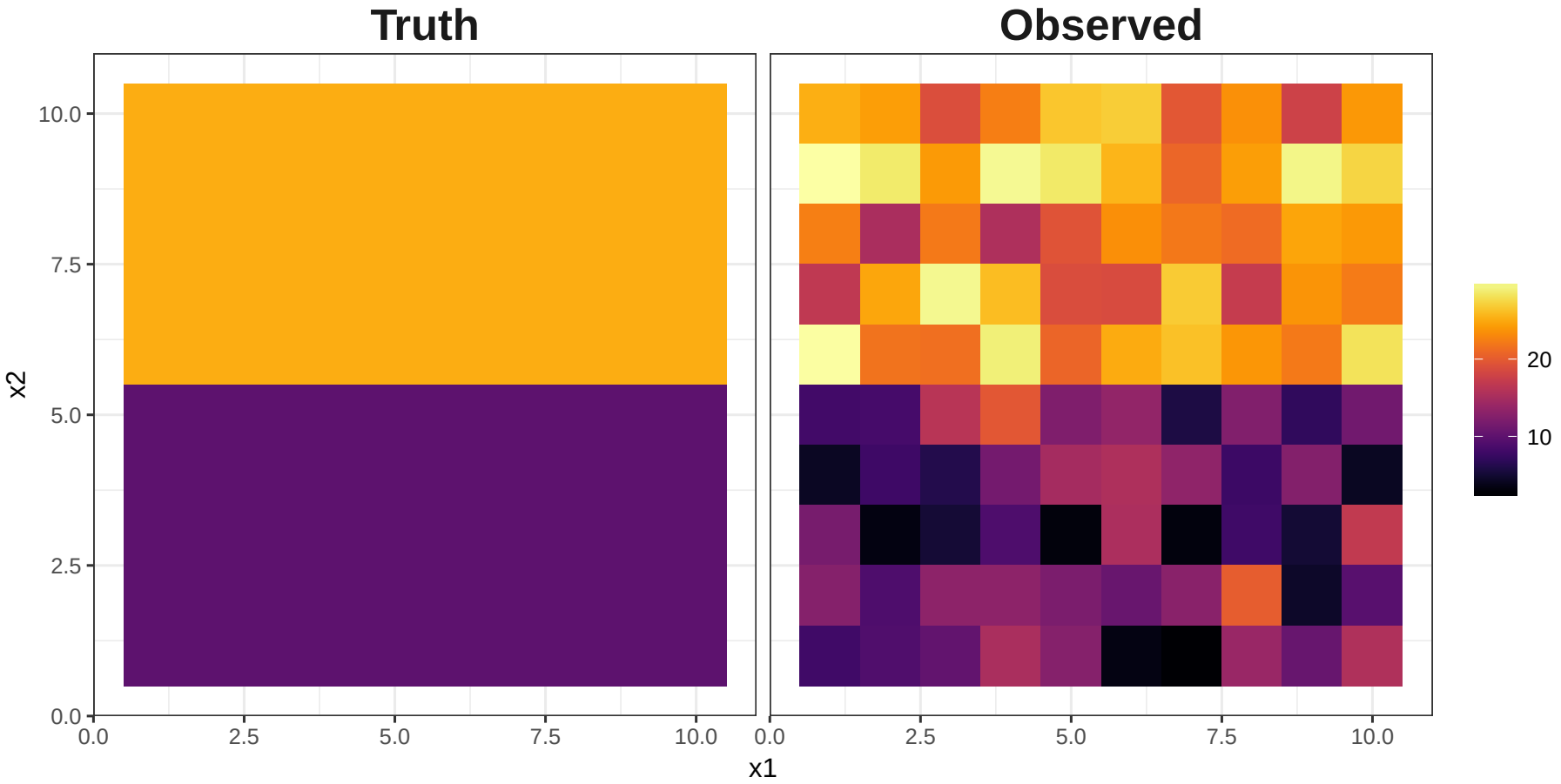
Simulated spatial processes

- Two scenarios
- 10 x 10 uniform grid
- Single observation per grid cell
- Observations are conditionally independent where

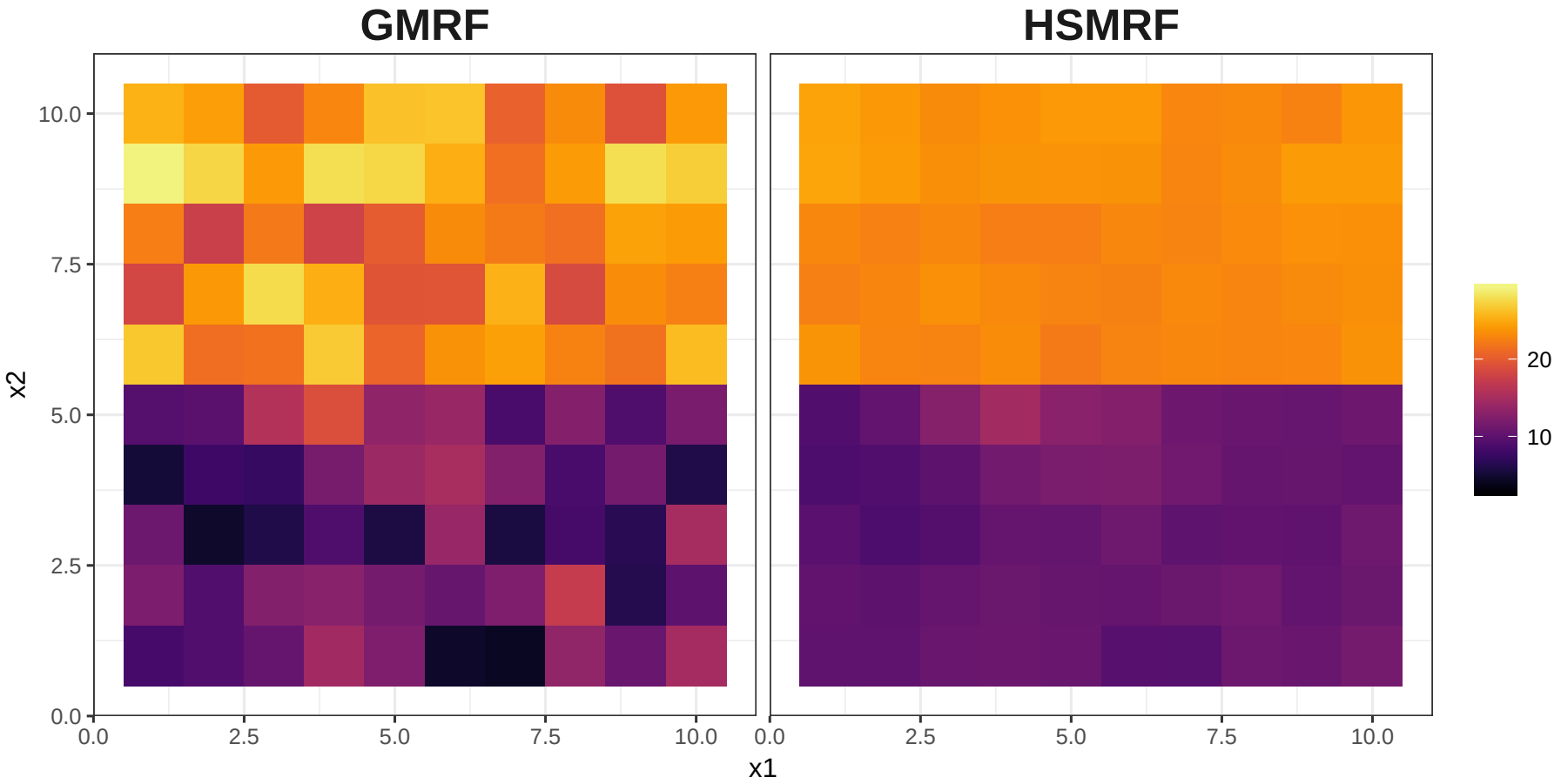
$$y_i \mid \boldsymbol{\theta} \sim \mathrm{N}(\theta_i, \sigma^2)$$

and $\sigma^2 = 4$ in each scenario

Simulated process 1



Simulated process 1 – Results



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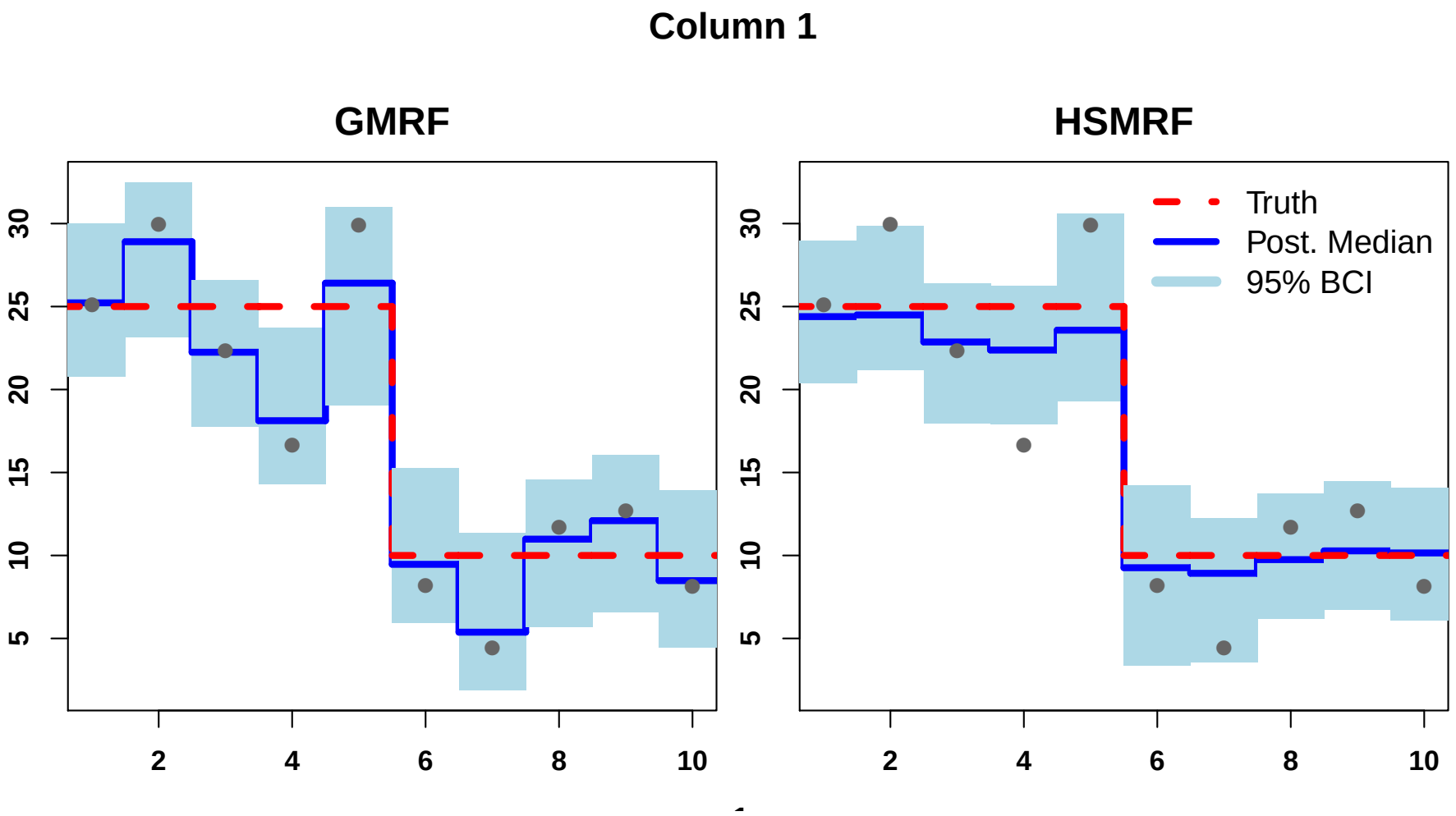

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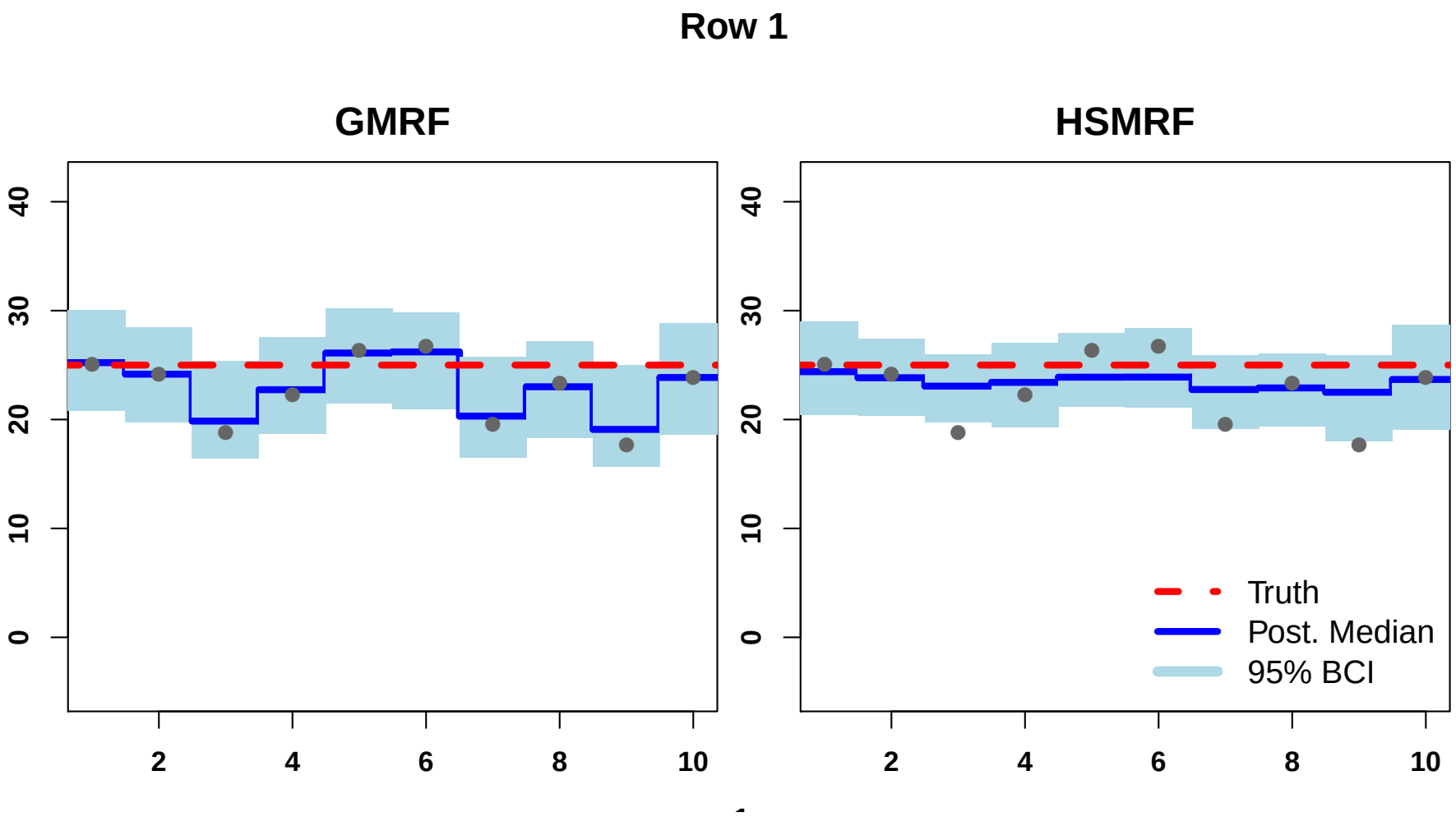
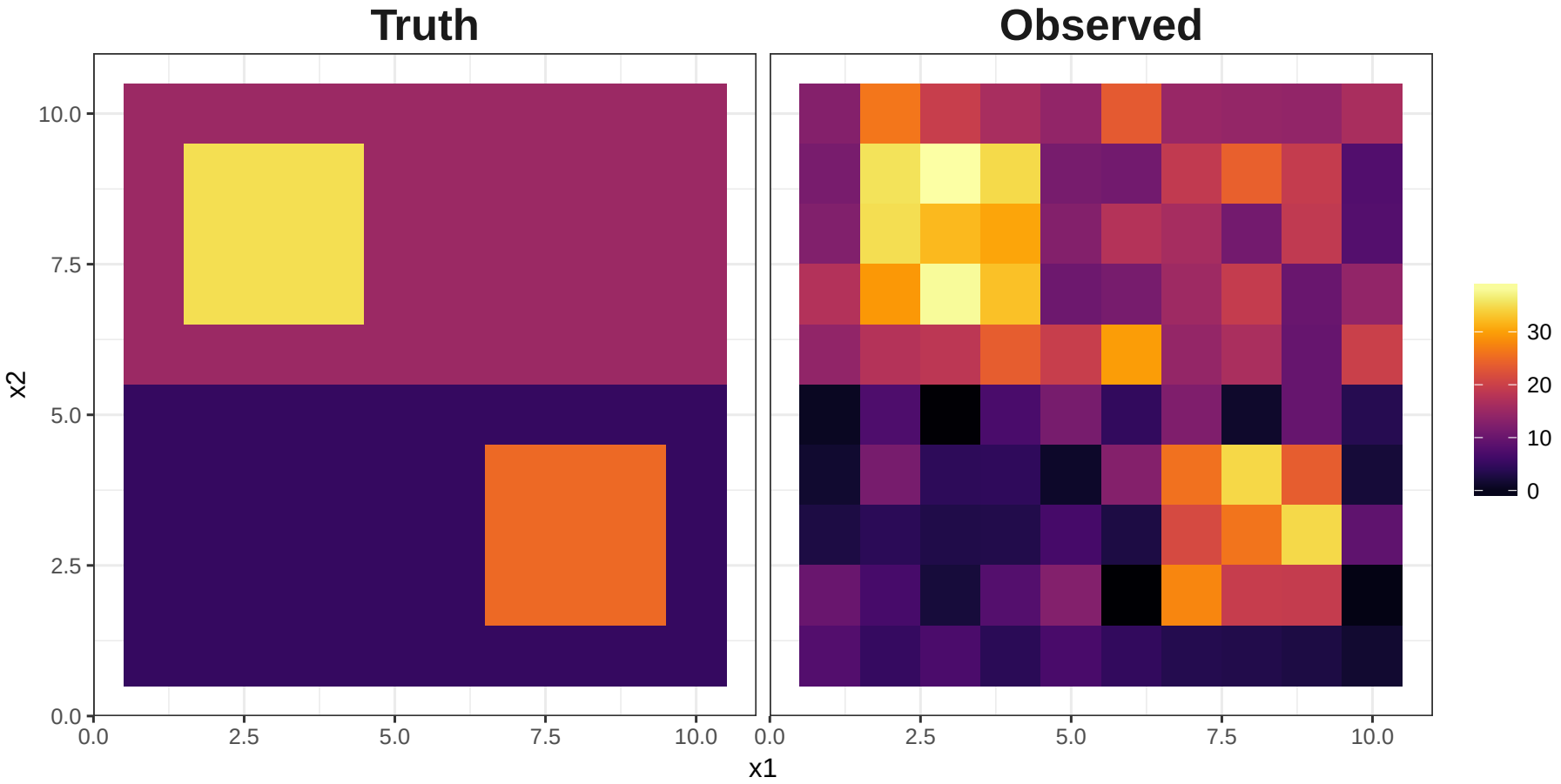
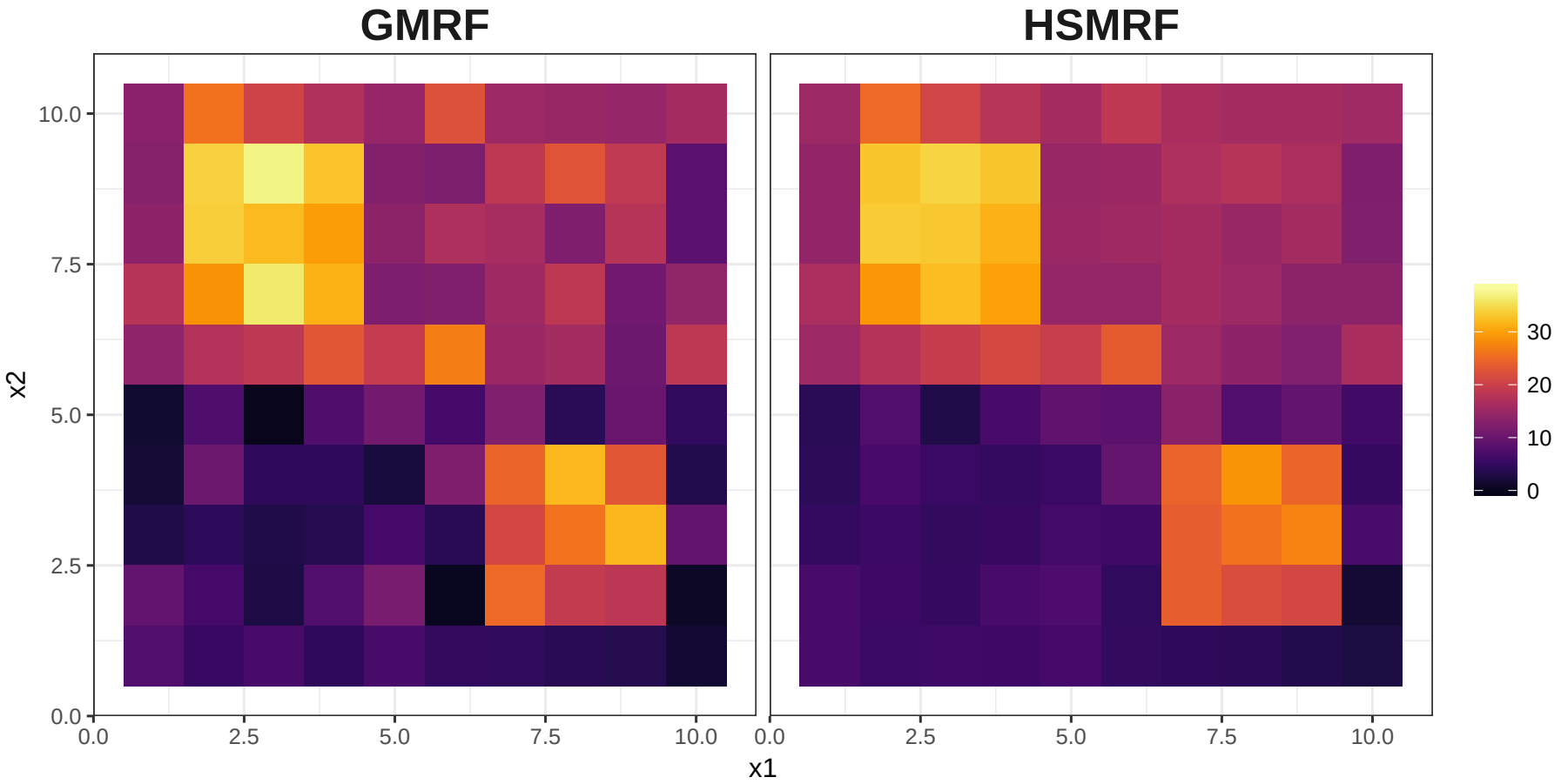


Simulated process 2



Simulated process 2 – Results



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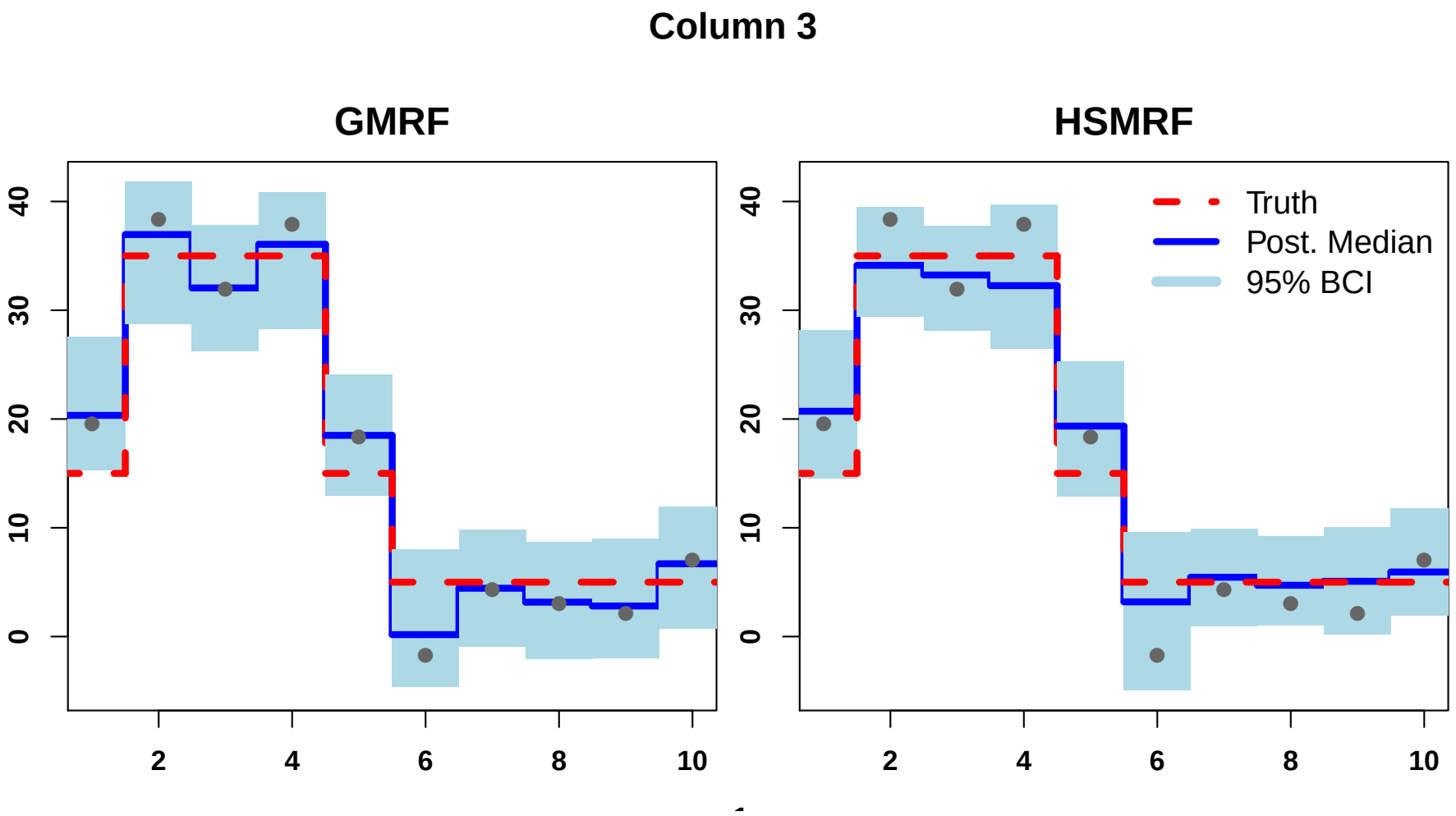

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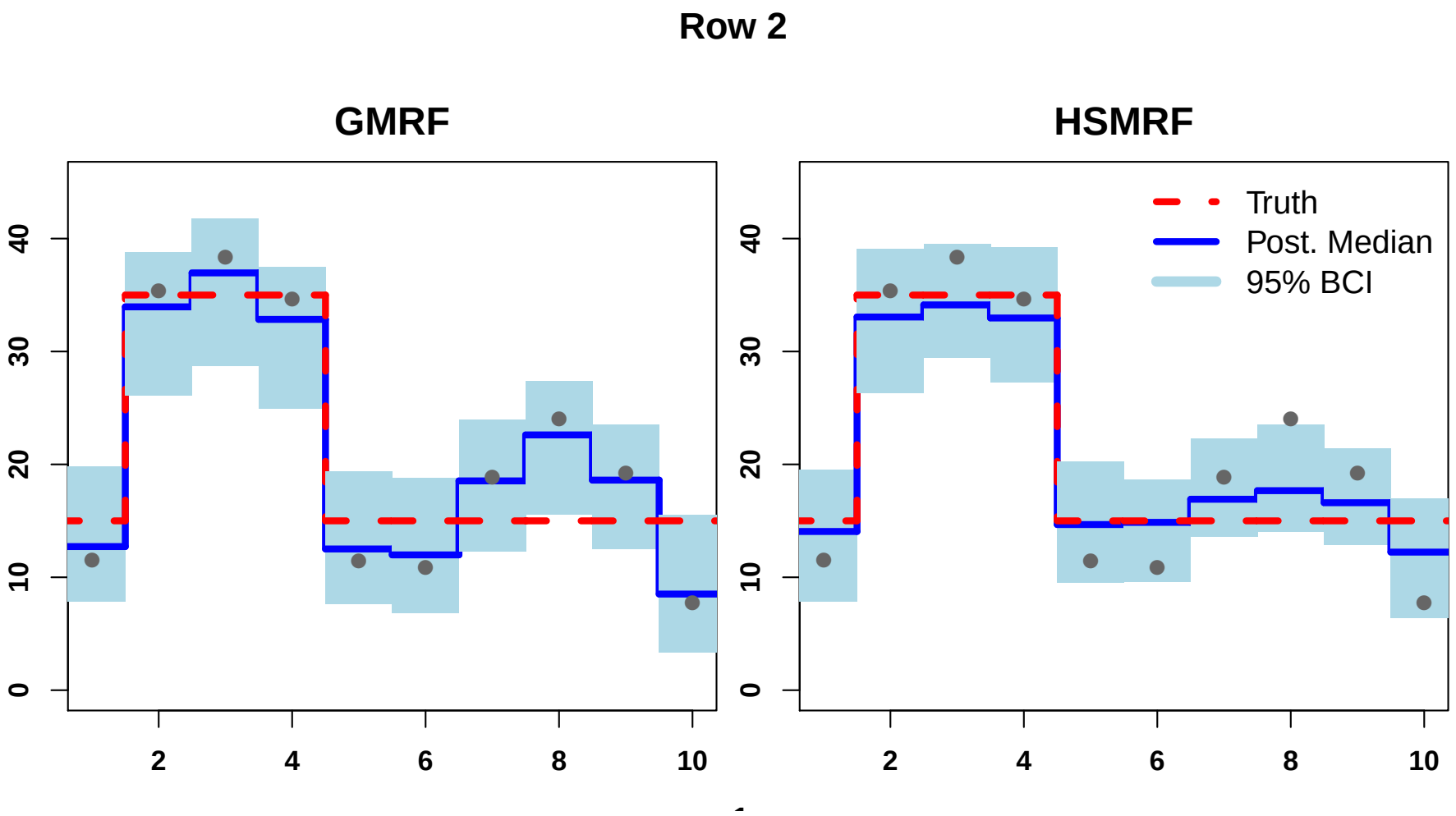
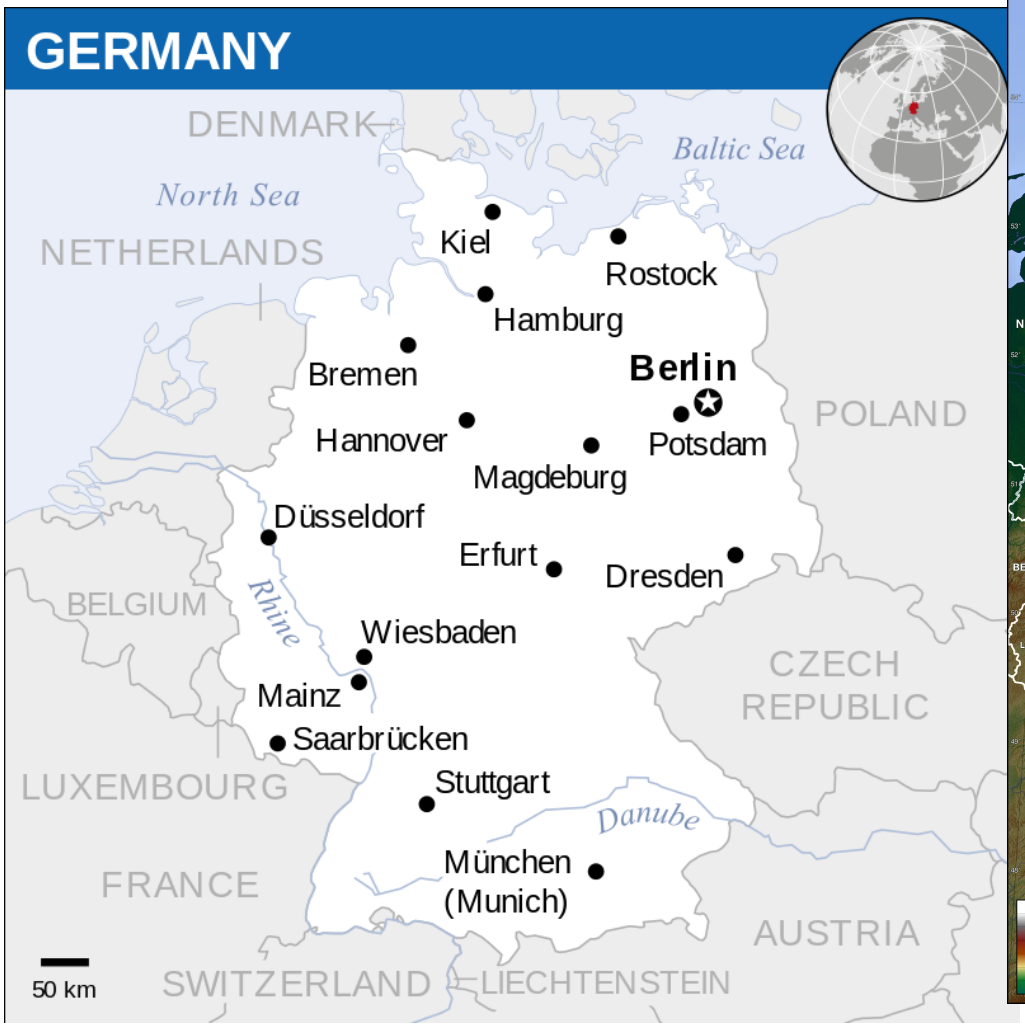


Data example: oral cancer in Germany

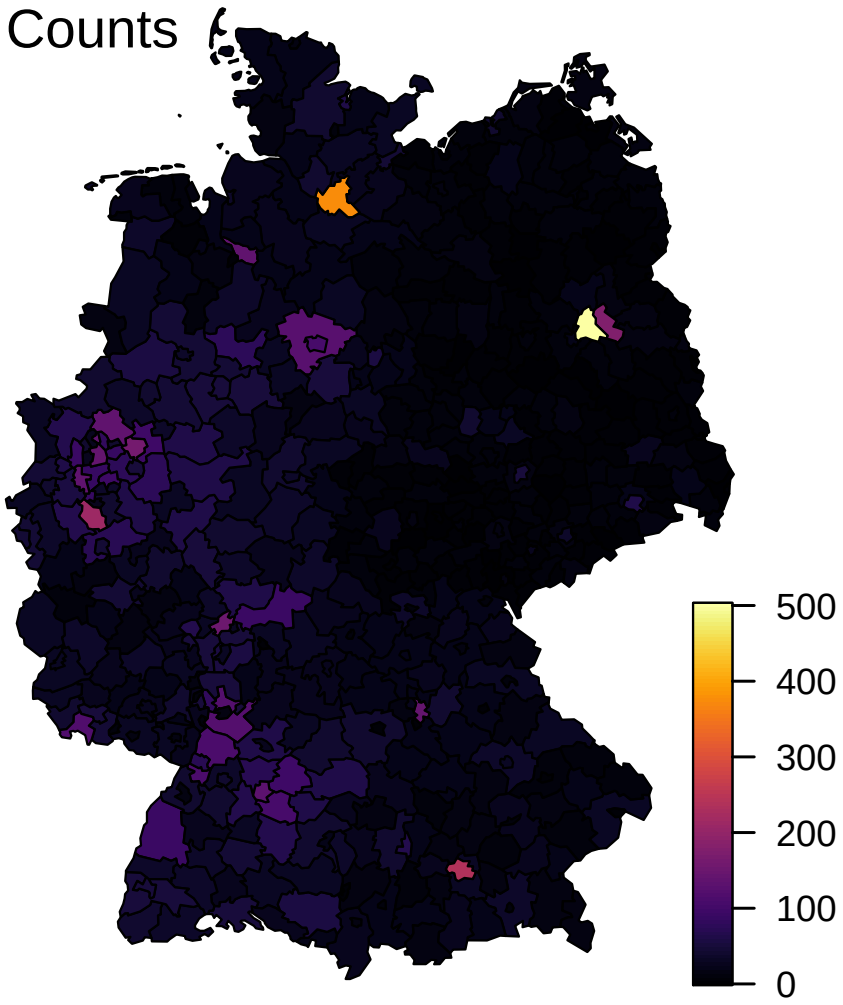
- Data are number of deaths from oral cavity cancer in males in Germany
- Also includes expected number of deaths based on age and population size
- Covers years 1986-1990 for 544 districts
- Total of 15,466 deaths, with median per district of 19 and range 1 - 501
- Interest in estimating relative risk of oral cancer spatially

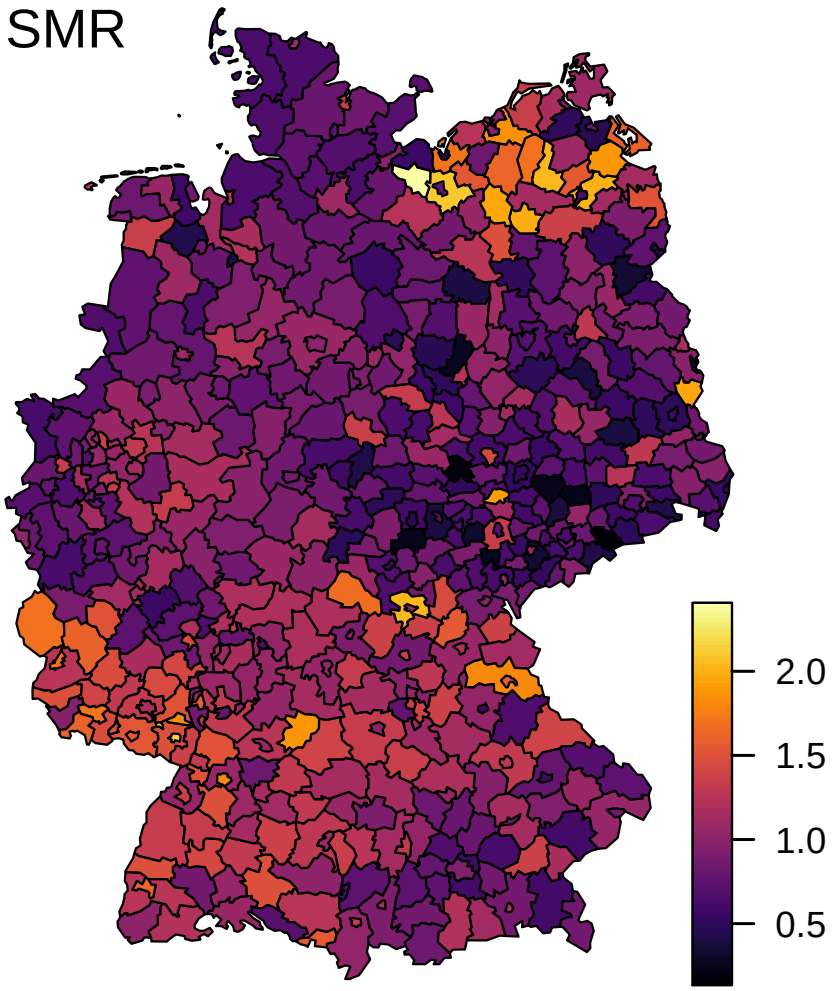
Germany


GERMANY
DENMARK
Baltic Sea
North Sea
NETHERLANDS
Kiel
Rostock
Hamburg
Bremen
Berlin
Hannover
Potsdam
Magdeburg
POLAND
Düsseldorf
Erfurt
Dresden
BELGIUM
Rhine
Wiesbaden
Mainz
CZECH REPUBLIC
Saarbrücken
LUXEMBOURG
Stuttgart
Danube
München (Munich)
FRANCE
AUSTRIA
50 km
SWITZERLAND
LIECHTENSTEIN

DÄNEMARK
Nordsee
Ostsee
Kiel
SCHLESWIG-HOLSTEIN
Rostock
Lübeck
Hamburg
Schwerin
MECKLENBURG-VORPOMMERN
Bremerhaven
Bremen
Wittstock
NIEDERSACHSEN
Hannover
BERLIN
Potsdam
Frankfurt
POLEN
NIEDERLANDE
BRANDENBURG
Magdeburg
NORDRHEIN-WESTFALEN
SACHSEN-ANHALT
Cottbus
Dortmund
Düsseldorf
Leipzig
Köln
Eifel
Zwickau
BELGIEN
Wiesbaden
Frankfurt
Mainz
PRAG
TSCHECHIEN
LUXEMBURG
Saarbrücken
BAYERN
FRANKREICH
BADEN-WÜRTTEMBERG
Augsburg
München
Freiburg
Konstanz
ÖSTERREICH
SCHWEIZ
LIECHTENSTEIN
km

Oral cancer data





Model 1 - Besag-type model

- Let y_i be the number of cancer deaths and N_i be the expected number in district i over the time period. Then we assume

$$y_i \mid \boldsymbol{\theta} \sim \mathrm{Poisson}(\exp(\theta_i)N_i)$$

- GMRF model:

$$\boldsymbol{\theta} \mid \gamma \sim \mathrm{N}(\mu, \boldsymbol{Q}^{-1})$$

- HSMRF model:

$$\boldsymbol{\theta} \mid \gamma, \boldsymbol{\tau} \sim \mathrm{N}(\mu, \boldsymbol{Q}(\boldsymbol{\tau})^{-1})$$

$$\tau_j \mid \gamma \sim \mathrm{C}^+(0, \gamma)$$

- For both models $\gamma \sim \mathrm{C}^+(0, 0.1)$

Model 1 - Besag-York-Mollie-type model

- Add independent random effects u_i for each district, such that

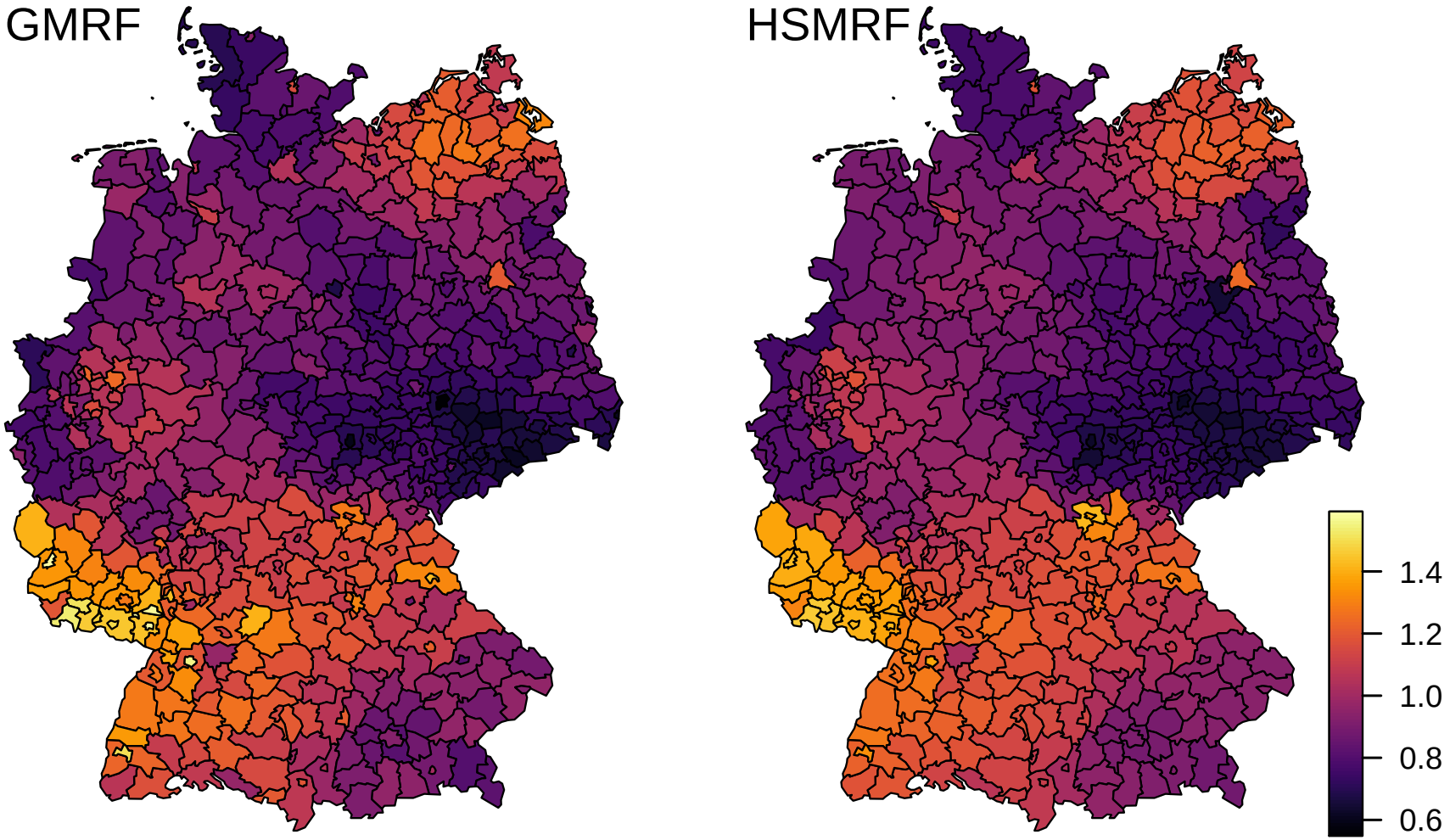
$$y_i \mid \boldsymbol{\theta} \sim \mathrm{Poisson}(\exp(\theta_i + u_i)N_i)$$

$$u_i \sim \mathrm{N}(0, \sigma^2)$$

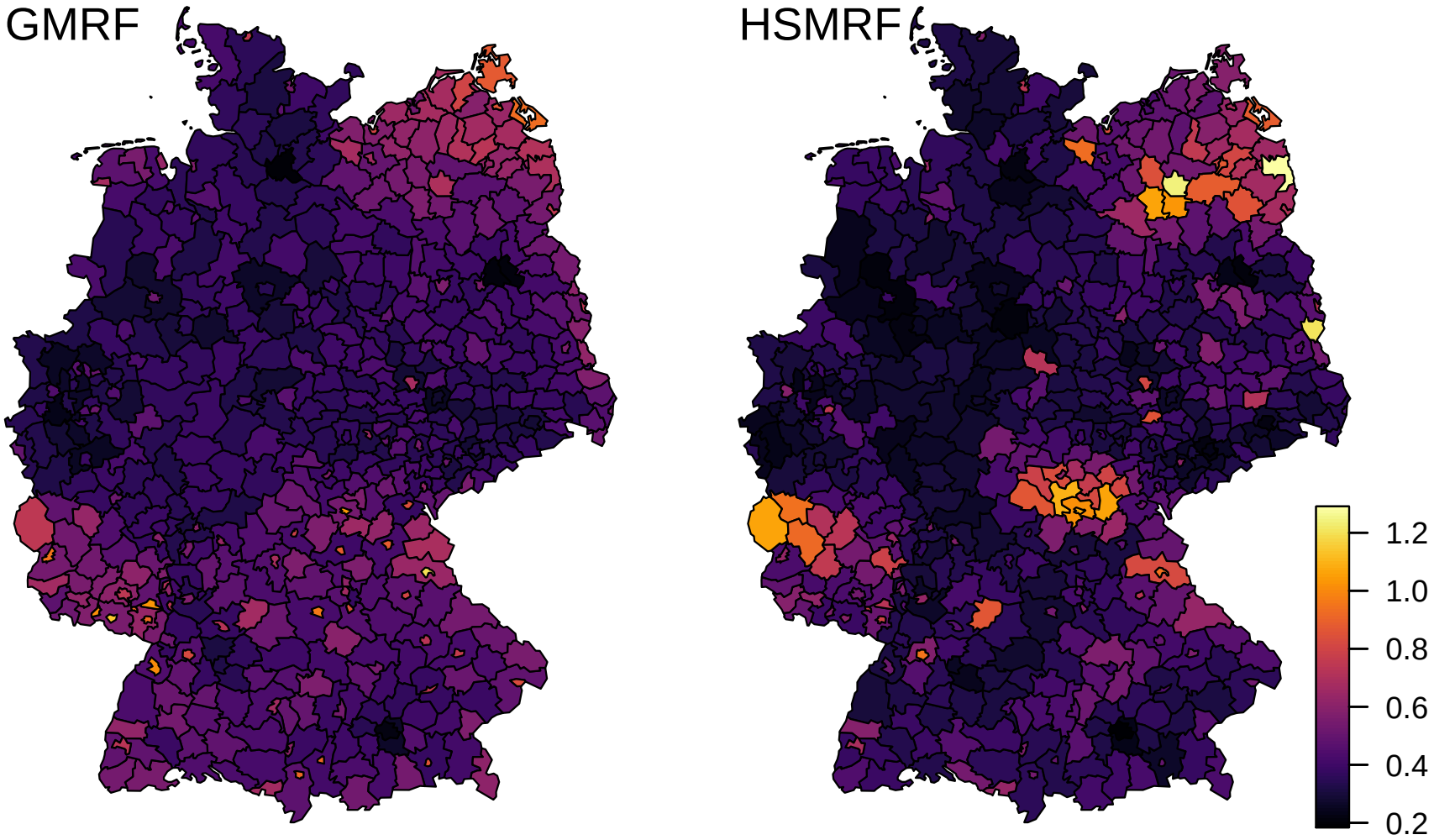
$$\sigma \sim \mathrm{C}^+(0, 5)$$

The priors for $\boldsymbol{\theta}$ stay the same.
For both models, interest is in surface for the relative risk, $\exp(\boldsymbol{\theta})$

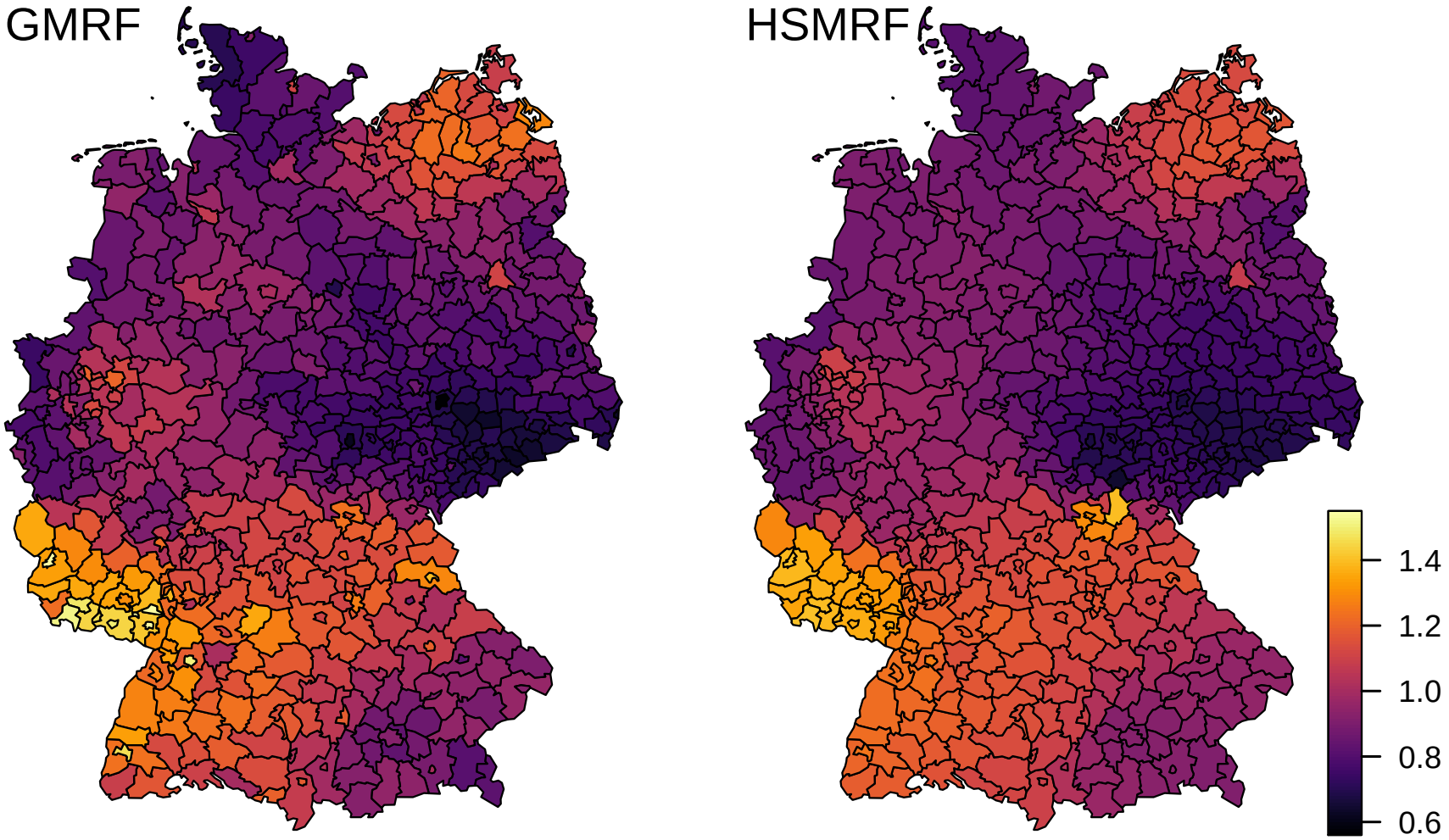
Posterior medians - Besag



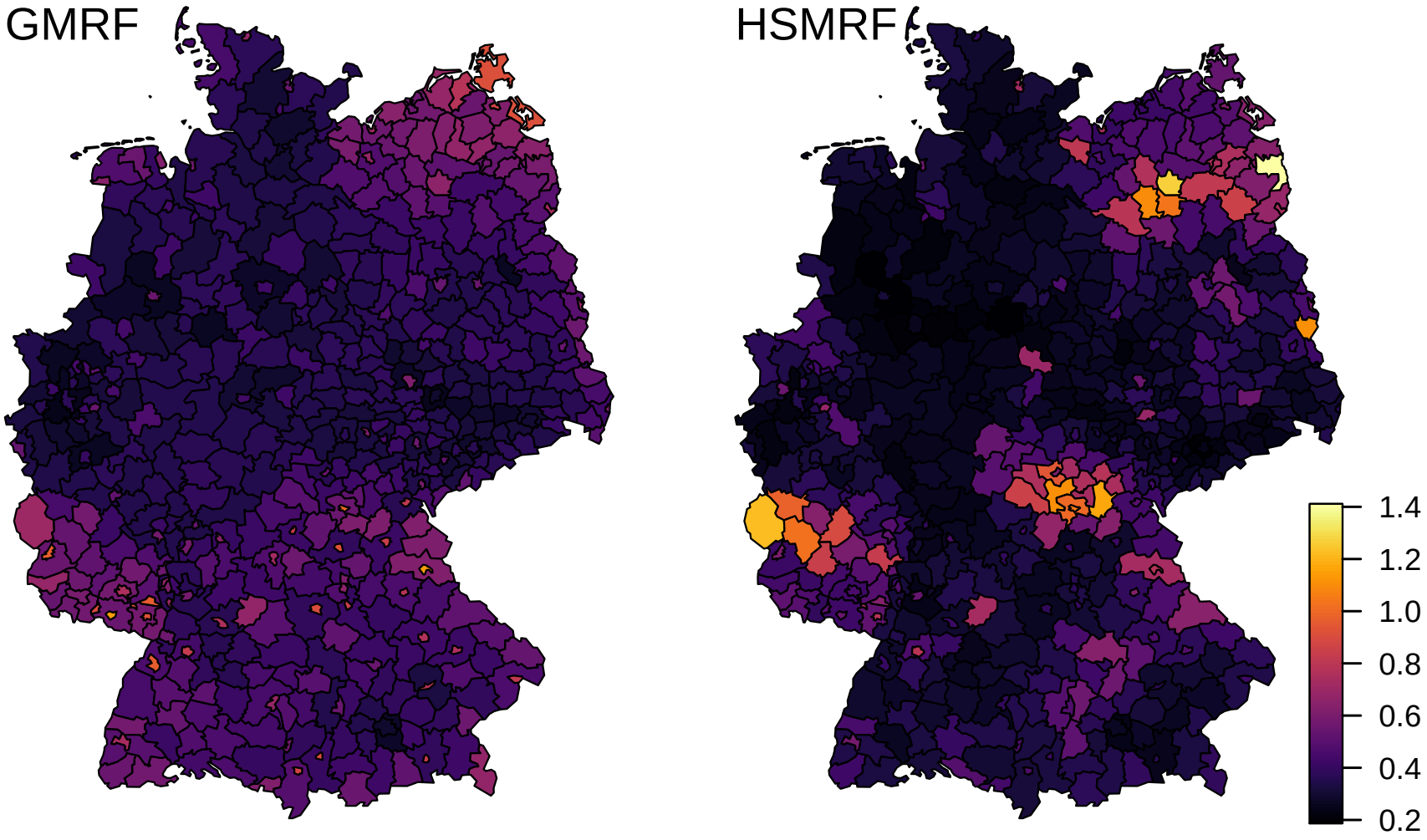
CI width - Besag



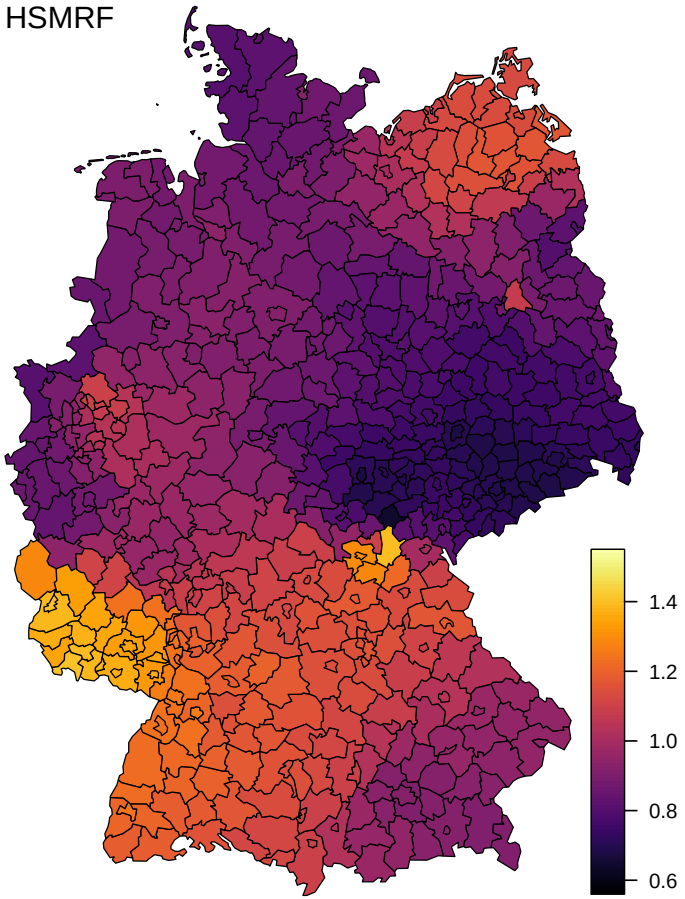
Posterior medians - BYM



CI width - BYM



States in Germany



Next steps

- Compare to other methods with simulations
- Allow addition of covariates
- Investigate how models respond to spatial confounding issues (e.g., Hughes and Haran 2013)
- Find other data examples

References

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